Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/29/2020
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	10/08/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 11/7/19) NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	80NSSC20M0119 ; NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station (ISS). The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows: 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	We compared the calculated convective heat flux, flame height, and radius, from the ellipsoidal model, with the ISS measurements. These are documented in a summary spreadsheet for the round 1 tests. We organized the existing ISS Round 1 data and also added to it to develop our “ISS summary spreadsheet,” which contains a summary of measurements and derived quantities. We compared measured heat flux at the center of the burner surface, flame height, and radii with their theoretical counterparts. We extracted steady heat flux on the burner surface and flame radiation measurements analysis to report heats of gasification (which corresponds to a range of solids) and radiative loss fraction correlation with flame height. We began an analysis of the video records from all four cameras in Round 1. We developed a Matlab code to read an image sequence with an unlimited number of images and to measure flame dimensions for each image. We worked on radiation fraction correlations with a non-dimensional theoretical variable and corrected radiative heat flux to the bevel edge, which was used to also correct the calorimeter for the bevel edge effect. We finalized the radiation dimensional analysis for prediction of flame radiative loss fraction. We are currently working on the uncertainty analysis. We worked on the ISS test matrix and documentation for the second round of testing (i.e., parameters spreadsheet, ignition flow timing calculations, and estimation of the heat-flux/flame-size steadiness to obtain an appropriate burn time for the experiments with the 50 mm burner). We performed RadCal calculations to determine absorption coefficients for varying oxygen mole fractions and fuel fractions for PMMA, POM, PP, and PS. Two test days have been completed in round 2. In the first day the flow rates specified in the test parameters were not achieved. This has now been corrected. We have requested a make-up day, which will be day 9 if the request is granted. We have made good progress analyzing the Day 1 and 2 results. We are converting the Advanced Combustion via Microgravity Experiments (ACME) camera still images into color videos. Predictions made for these tests are being compared with the measured results. We have continuously updated the Round 2 test matrix and parameters as tests are performed. This includes changing the test day order and the fuel bottles and flow meters associated with each day.
Bibliography Type:	Description: (Last Updated: 10/08/2025)

Articles in Peer-reviewed Journals	Auth E, Quintiere JG, Sunderland PB. "Emulation of condensed fuel flames with gaseous fuels supplied through a porous copper calorimeter." Fire and Materials. 2020 Nov;44(7):935–42. https://doi.org/10.1002/fam.2896 , Nov-2020
Papers from Meeting Proceedings	Dehghani P, Auth E, Cui C, Stocker DP, deRis JL, Sunderland PB, Quintiere JG. "Microgravity observations of a burning emulator to investigate material flammability." 2020 Spring Technical Meeting of the Eastern States Section of the Combustion Institute, Columbia, SC, March 8-11, 2020. Abstracts. 2020 Spring Technical Meeting of the Eastern States Section of the Combustion Institute, Columbia, SC, March 8-11, 2020. , Mar-2020